Dilan Karaguler

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Research Interests

Applied algebraic topology and topological data analysis; path homology and Mayer (N-nilpotent) homology of directed graphs and path complexes; persistent homology; topology-informed graph machine learning and geometric deep learning, with applications to molecular, dynamical and engineering data.

Education

Michigan State University — East Lansing, MI, US
Ph.D. in Mathematics (Candidate) — August 2022 – May 2027 (Expected)
- Advisors: Prof. Ekaterina Rapinchuk and Prof. Guowei Wei

Middle East Technical University (METU) — Ankara, Türkiye
M.S. in Mathematics — September 2019 – September 2021
- Thesis: *A Survey on Multidimensional Persistence Theory*
- Advisor: Assoc. Prof. Semra Pamuk

Middle East Technical University (METU) — Ankara, Türkiye
B.S. in Mathematics, Honor Student — June 2014 – June 2019

Publications

D. Karaguler and G.-W. Wei. *Mayer Path Homology.* Foundations of Data Science, accepted (2026). doi:10.3934/fods.2026025, arXiv:2605.16525.

D. Karaguler. *A Survey on Multidimensional Persistence Theory.* M.S. Thesis, Middle East Technical University (2021).

D. Karaguler and G.-W. Wei. *Persistent Mayer Path Laplacians.* In preparation (2026).

Research Experience

Graduate Researcher — August 2022 – Present
Michigan State University — East Lansing, MI, US
- Develop Mayer path homology, a homology theory for directed graphs built from N-nilpotent differentials on path complexes; published in *Foundations of Data Science* (with G.-W. Wei).
- Implement topological feature extraction (path complexes, Mayer homology and Laplacians, persistent homology) in Python with GUDHI, NumPy and PyTorch, and combine these features with graph neural networks.

Research Mentor — May 2023 – June 2023
Topology RTG, Michigan State University — East Lansing, MI, US
- Mentored an undergraduate researcher on a project in Morse theory: project design, methodology and research presentation; the student went on to major in mathematics.

Industry Experience

Software Engineer Intern — May 2026 – August 2026
Intel Corporation — Oregon, US
- Built an end-to-end graph transformation pipeline in Python that extracts schematic topology directly from chip layout files, without pre-existing schematic metadata.
- Modeled low-level device instances as graph nodes and extracted a local subgraph around each instance to capture its connectivity.
- Developed graph aggregation and embedding routines producing a vector representation per device, used to classify components as analog, digital or fill devices.
- Worked with Design Yield and Manufacturability engineers to evaluate the prototype for automated structural recognition in layout verification.

Graduate Technical Intern — May 2025 – August 2025
Intel Corporation — Oregon, US
- Performed spatial pattern analysis on semiconductor chip layer images with KLayout and Python to identify and cluster recurring structures across layout datasets.
- Applied graph neural networks to search for and classify similar patterns across multi-layer designs.
- Partnered with cross-functional yield teams to bring ML-based pattern recognition into automated yield-analysis workflows.

Teaching Leadership & Mentoring

Lead Teaching Assistant August 2024 – Present
Department of Mathematics, Michigan State University East Lansing, MI, US

- **2026 – 2027:** Co-lead, with Eloy Moreno-Nadales, a yearlong workshop series for first-year graduate TAs: introductions, *Three Aspects of Teaching*, mock teaching with peer feedback, and a graduate student panel.
- **Fall 2025 (MTH 132):** Prepared recitation worksheets, gathered TA feedback for instructors, and substituted for recitation leaders.
- **Spring 2025 (MTH 133):** Ran TA meetings, prepared weekly worksheets, and set up Teams tools to track grading deadlines.
- **Fall 2024:** Ran ULA orientation and mock-teaching sessions, and observed ULAs and GTAs to give feedback.

Graduate Lead, Center for Instructional Mentoring (CIM) Summer 2024
Department of Mathematics, Michigan State University East Lansing, MI, US

- Mentored first-time graduate instructors through weekly check-ins and non-evaluative feedback.
- Created a teaching handbook from interviews with senior GTAs on each course's expectations.

Mentor, Academic Culture and English Skills (ACES) Program Summer 2024
Michigan State University East Lansing, MI, US

- Mentored incoming international graduate TAs on U.S. classroom culture and cross-cultural communication; organized social events.

Supervisor, Math Learning Center (MLC) Fall 2022 – Spring 2023, Fall 2024
Michigan State University (Teaching Exemption Fellowship, 2022 – 2023) East Lansing, MI, US

- Managed how tutors were distributed among students and arranged the room to match demand.

Student feedback: dilankaraguler.github.io/teaching/what-students-say

Courses Taught

MTH 314 — Matrix Algebra with Computational Applications Fall 2026
Instructor (1 section), Michigan State University East Lansing, MI, US

- Prepare and deliver lectures, facilitate group work, and hold help-room hours.

MTH 299 — Transitions Fall 2025
Teaching Assistant (2 sections), Michigan State University East Lansing, MI, US

- Co-taught with Ester Sgallova: weekly recitations, homework solutions, grading and help-room hours.

MTH 314 — Matrix Algebra with Computational Applications Spring 2025
Teaching Assistant (1 section), Michigan State University East Lansing, MI, US

- Graded homework and the final exam; ran quiz and exam reviews and help-room hours.

MTH 133 — Calculus II Fall 2024
Teaching Assistant (1 section) and pod leader, Michigan State University East Lansing, MI, US

- Led a recitation; as pod leader, ran recitation-leader meetings and wrote the weekly worksheets.

MTH 235 — Differential Equations Spring 2024
Teaching Assistant (3 sections), Michigan State University East Lansing, MI, US

- Wrote recitation notes with summaries and practice problems; graded quizzes and held help-room hours.

MTH 133 — Calculus II Fall 2023
Teaching Assistant (3 sections: 2 online, 1 in person), Michigan State University East Lansing, MI, US

- Prepared weekly lecture summaries; graded and proctored quizzes; held help-room hours and followed up with online students.

MTH 133 — Calculus II Summer 2023
Teaching Assistant (1 online asynchronous section), Michigan State University East Lansing, MI, US

- Contacted students weekly by email and tracked their course activity.

MTH 154, 153, 120, 119 — Calculus I and II January 2020 – June 2022
Graduate Teaching Assistant, Middle East Technical University Ankara, Türkiye

- MTH 154 (Spring 2022) and MTH 153 (Fall 2021) for Mathematics Students; MTH 120 (Spring 2021) and MTH 119 (Fall 2020) for Engineering Students.
- Led recitations and office hours; graded and proctored exams.

MTH 154 — Calculus II for Mathematics Students Spring 2019
Undergraduate Teaching Assistant (homework grader), Middle East Technical University Ankara, Türkiye

- Graded homework for more than 300 students.

MTH 153 — Calculus I for Mathematics Students Fall 2018
Undergraduate Teaching Assistant (homework grader), Middle East Technical University Ankara, Türkiye

- Graded homework for more than 300 students.

MTH 116 — Introduction to Abstract Algebra Spring 2017
Undergraduate Teaching Assistant (recitation instructor), Middle East Technical University Ankara, Türkiye

- Conducted recitations for 180 students with my own recitation materials.

Curriculum Development

MTH 314 Matrix Algebra (Instructor): lectures and group-work activities — Fall 2026
MTH 132 Calculus I (Lead TA): weekly group activities with Desmos explorations — Fall 2025
MTH 299 Transitions: homework solutions as a recitation leader — Fall 2025
MTH 133 Calculus II (pod leader, Lead TA): weekly recitation worksheets for all sections — Fall 2024 – Spring 2025
Teaching handbook for first-time graduate instructors, Center for Instructional Mentoring — Summer 2024
MTH 235 Differential Equations: recitation notes with summaries and practice problems — Spring 2024
MTH 133 Calculus II: weekly lecture summaries for recall sessions — Fall 2023
METU MTH 154 Calculus II: recitation questions and homework assignments — Spring 2022
METU MTH 153 Calculus I: recitation questions and homework assignments — Fall 2021
METU MTH 116 Introduction to Abstract Algebra: recitation materials — Spring 2017

Teaching Development & Educational Research

Program Evaluator & Instructional Researcher — May 2026
AI for Advanced Semiconductor Manufacturing Workshop, Michigan State University — East Lansing, MI, US
- Designed pre/post surveys and observation protocols, and analyzed feedback with quantitative and thematic coding.
- Wrote an evaluation report with curriculum recommendations grounded in adult learning theory.

MTH 879 Teaching College Mathematics (graduate course) — Fall 2024
Michigan State University — East Lansing, MI, US

Colleges Online Learning Academy (COLA) Scholar and workshops — Summer 2024
Michigan State University — Portfolio
- Completed workshops on online course design; built a teaching portfolio website and an online course plan.

Certification in College Teaching Institute (CCTI), two-day workshop — May 2024
Michigan State University — East Lansing, MI, US

Talks

Integrating AI to Lecture Prep (upcoming) — November 2026
Mathematics Teaching Seminar, *Sparks: Teaching Lightning Talks by Graduate Instructors* — MSU

Mayer Path Homology — April 2026
TDA Seminar, Dept. of Computational Mathematics, Science and Engineering, MSU — East Lansing, MI, US

Interaction Topology in TDA — October 2025
Invited lecture, MTH 994: Mathematics-Assisted AI and AI-inspired Mathematics, MSU — East Lansing, MI, US

Mayer Topology in TDA — October 2025
Invited lecture, MTH 994: Mathematics-Assisted AI and AI-inspired Mathematics, MSU — East Lansing, MI, US

From Data to Barcodes — April 2021
Graduate Student Seminar, Middle East Technical University — Ankara, Türkiye

Poster Presentations

Mayer Path Topology and Persistence — September 2025
SIAM Great Lakes Section Meeting, Illinois Institute of Technology — Chicago, IL, US

Mayer Path Topology and Persistence — March 2025
Mathematics of Data, Dynamics, and Life Sciences Conference, UC Irvine — Irvine, CA, US

Projects

Persistent Mayer Homology with Dynamic Mode Decomposition — April 2026
Python, NumPy, SciPy, GUDHI — GitHub
- Built a reproducible pipeline combining Vietoris–Rips filtrations, Mayer homology and Dynamic Mode Decomposition (DMD) to track time-varying topological features.
- Computed Mayer Betti trajectories on synthetic 3D point dynamics; a rank-10 DMD model reconstructed a 500-frame dataset with about 3.6% relative error and supported multi-step forecasting.

Homology-Fed Neural Network Framework — November 2024
Python, PyTorch, GUDHI, NumPy — GitHub
- Wrote a Python library computing Mayer homology and path-complex Laplacians on graph-structured data.
- Used the resulting topological features for molecular property prediction on the QM9 dataset with graph neural networks.

Cellular Automata on Curved Surfaces — March 2024
Python, NumPy, Matplotlib — GitHub
- Simulated Conway's Game of Life on a square, torus and sphere with surface-specific neighbor rules, and analyzed how board size and initial density affect population persistence.

Awards, Fellowships & Scholarships

Doctoral Student Summer Internship Fellowship, The Graduate School, Michigan State University — Summer 2026
Excellence in Teaching Award, Department of Mathematics, Michigan State University — May 2025
Colleges Online Learning Academy (COLA) Fellowship, Michigan State University — Summer 2024
Teaching Exemption Fellowship, Michigan State University — 2022 – 2023
RONESANS Scholarship, RONESANS — 2018 – 2019
UNIGENC Scholarship, UNIGENC Association — 2015 – 2019
Merit Scholarship, Turkish Ministry of National Education — 2010 – 2014

Service, Leadership & Outreach

Graduate Representative, NatSci Student Advisory Council (SAC) — 2025 – 2026
College of Natural Science, Michigan State University — East Lansing, MI, US

Graduate Student Representative, Graduate Studies Committee — 2024 – 2025
Department of Mathematics, Michigan State University — East Lansing, MI, US
- Represented mathematics graduate students on the committee and co-planned Responsible Conduct of Research (RCR) workshops.
- Hosted town halls, met prospective students, and directed students with concerns to department resources.

Member, First-Generation/Low-Income (FGLI) Group — September 2024 – May 2025
Department of Mathematics, Michigan State University — East Lansing, MI, US
- Took part in community gatherings and social events with first-generation and low-income undergraduates.

Organizing Committee Member — April 2024
Graduate Student Topology and Geometry Conference, Michigan State University — East Lansing, MI, US
- Handled speaker communication, scheduling and room assignments; moderated sessions.

President / Board Member — August 2022 – August 2025
Association for Women in Mathematics (AWM), MSU Chapter — East Lansing, MI, US
- Organized seminars, career workshops, academic-life sessions and social events for women in mathematics.

President — February 2015 – December 2018
METU Math Club — Ankara, Türkiye
- Organized seminars, workshops and reading groups; prepared first-year students to lead abstract algebra and number theory discussions.

Outreach & Volunteering

Proctor, Math Kangaroo Team Competition, Department of Mathematics, MSU (upcoming) — November 2026
Volunteer, Math Night, Marble Elementary School, East Lansing (upcoming) — November 2026
Volunteer, MSU Student Food Bank, Michigan State University — Spring 2024

Conferences & Workshops Attended

Agentic AI (hot topic course), AIDMM NSF Research Traineeship, Michigan State University — Fall 2026
AI, Data and PDE Workshop, Michigan State University — April 2025
Graduate Student Topology and Geometry Conference, Michigan State University — April 2024
Mid-Atlantic Topology Conference, Northeastern University — March 2024
SIAM Great Lakes Section Meeting, Michigan State University — October 2023
Computational Persistence, Purdue University — September 2023
Randomness in Topology and its Applications, IMSI, Chicago — March 2023
LMS Workshop: Applied Algebraic Topology, Queen Mary University of London (online) — February 2022
Topological Insights in Neuroscience, IMSI (online) — May 2021
Topological Data Analysis, IMSI (online) — April 2021

Technical Skills

Programming: Python, R, SQL
Methods: Graph Neural Networks, Graph Transformers, Topological Data Analysis, Persistent Homology, Vietoris–Rips Filtrations, Dynamic Mode Decomposition, Supervised/Unsupervised Learning, Model Evaluation
Libraries: PyTorch, PyTorch Geometric, NetworkX, GUDHI, NumPy, SciPy, Pandas, Scikit-Learn, TensorFlow, Keras, Matplotlib, XGBoost, LightGBM
Tools: Git, Linux/Unix, DuckDB, SQLite, Jupyter, KLayout, LaTeX
Domain Knowledge: Design for Manufacturability (DFM), Semiconductor Yield Analysis, Schematic Extraction, Chip Layout Analysis

References

Available upon request.